

Envisioning Digital Sanctuaries: An Exploration of Virtual Collectives for Nurturing Professional Development of Women in Technical Domains

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ABSTRACT

Work and learning are essential facets of our existence, yet socio-cultural barriers may limit access and opportunity in such contexts. Historically, women have faced multiple restrictions that have hindered their entry into professional ventures, many of which have also plagued the technical domains of Information technology and Computer science. Hence, it is essential to investigate how women can subvert such structural subjugation and can find channels through which they can seek support, and guidance and develop collective endurance to navigate their careers. In this regard, virtual communities (e.g., Reddit) have emerged as crucial spaces that allow community building, and opportunities for mobilizing collective action and have also emerged as sanctuaries for support and companionship. Thus, this proposed dissertation aims to focus on how these spaces are used and adopted by women for resilience building and navigating goals, aspirations, and tensions in crafting their professional journey in technical domains. The layers of analysis will include understanding the present conversational patterns in three subreddits dedicated to professional development, and design exploration to better unpack how such channels can be envisioned in the future to more deeply align the affordances of such channels with the values of these virtual spaces. The long-term goal for this work is to address the way in which such channels can be designed and curated to offer spaces for enrichment, empowerment, and advocacy with a focus on professional development for women, especially for those engaged in technical domains.

KEYWORDS

Resilience building and support; Online community norms and practice; Gender in the workplace; Content analysis; Network analysis; Reddit

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1 INTRODUCTION

The overarching aim of my dissertation work is to understand the use of online discussion channels for supporting professional development of women especially in technical fields such as Computer Science and Information technology. Particularly, I wish to address the broader question of how such online channels should be designed and structured in order to boost the overall collaborative efficacy of such spaces while also offering a safe, fair, open, and equitable space for extended collaboration and deliberation. Work, learning, and establishing one's professional footprint can be integral to our day-to-day lives. However, access and ability to sustain one's professional footprint might be contingent upon one's social standing and identity [15].

Women have historically have faced multiple such socio-cultural limiters that have impeded their growth and ability to attain fulfillment in professional ventures [15]. In this regard, social constructivist frameworks have surmised how the construct of gender emerges in practice. Pertinent in this context is Pierre Bourdieu's conceptualization of the gendered habitus that highlights how a structure emerges around the construct of gender that affects the agency of women. Such structural subjugations are further magnified in the technical domains of information technology and computer science, which historically have had an organizational cultural tradition that has placed women and other marginalized groups at the periphery [11]. Essentially such forms of social isolation may result in a habitus of dominance that has been captured through concepts such as systems of exclusion manifesting in the everyday practices of such fields through contexts such as hiring practices, roles defined in organizational settings, and workplace cultures to name a few [11]. The presence of such structurally embedded exclusionary forces, may cause strife, turmoil and crisis in the everyday context which thus calls for the need to find channels and avenues for everyday resilience building [27]. Thus, crafting one's professional footprint may entail the need for not only having access to knowledge and information but also for giving and receiving extended communal support, mentoring, understanding, and care [12]. Thus, pivotal high level questions raised in this regard include: (1) *how can women overcome such socio-cultural constraints, particularly as they carve out professional identities in technical fields?* (2) *What tools and mediums can provide them with a platform to collectively resist, seek refuge and from peer led informal enclaves of resilience and support?*

To address these socio-cultural concerns, the goal of this dissertation project is to explore the viability and feasibility of using online communities as tools and socio-technical mediators, that

facilitate and nurture emergent community-building tactics in order developing a collective voice to empower underrepresented communities (such as women) [9]. In addition to acting as pivotal avenues for information and knowledge sharing [28], social media platforms in recent times have emerged as collectives of resilience, that offer a safe haven for those who might face multiple intersecting social hindrances and stigmas [20]. Several studies have indicated how such channels offer the opportunity to build community and provide multiple dimensions of support which can help to develop collective endurance [30]. The conversational practices and values found in such virtual collectives may serve as an example of a performative habitus that is created and sustained through the emergent community norms of reciprocity and exchange [21]. Yet, some studies also caution about how such collectives may perpetuate social stereotypes and introduce barriers of association for marginalized groups [26]. In this regard, empirical studies, drawing inspiration from the tenets of feminist HCI perspectives, have called for a deeper grounding of the values of such spaces along the intertwining axes of participation, inclusivity, quality of content, technical features (such as voting systems, tagging mechanisms) and governance [3]. These insights highlight that the design of such forums, acting as the key mediating variable, can often drive the way in which community members attach meaning and value along the various modalities of interaction or the affordances such forums offer [16, 34]. Further in the context of open online forums such as Wikipedia and Stack Overflow, prior studies have indicated that perceptions and inequitable norms of engagement deter participation and contribution from women [22]. Such inhibitors typically manifest as lack of peer parity, lack of recognition and respect for participation [13]. Such investigations indicate the need for deeper explorations into the way in which socio-technical values and affordances need to be possibly reassessed or can be etched into the technical fabric of such communities to empower historically suppressed voices such as women.

1.1 Theoretical motivation and related work

The theoretical grounding of this work is rooted in the sociological framework of the habitus, postulated by Pierre Bourdieu [5]. The key contribution of this line of thought lies in explicating how inequality is constructed and gets ingrained in the day-to-day practices of social life. Essential in this regard, is the notion of the habitus, the fact that an individual's values, beliefs, instincts and moral ideologies are shaped by and shape one's embeddedness or social position. Such a habitus may exist in the organizational context, that one may have to negotiate as they carve their professional identity, calling for the need to find avenues to network, seek knowledge and guidance for career advancement and identity development [7]. Further expanding this notion of the habitus and how it is entrenched in a spirit of inequality rises the notion of the gendered habitus. The gendered habitus, in Bourdieu's view is something that happens subconsciously and is so deeply etched into cultural practices that it becomes a routinized act that remains in certain sense unquestionable, appearing as if it is natural which results in a form of gendered social condition that Bourdieu terms as doing gender [12]. Yet, embedded in the gendered habitus are also notions and practices of social inequity, which may result in

acts of symbolic violence and domination on women even in the workplace context [24]. Especially in the context of the technical workplace, prior scholarship has pointed to how such socio-cultural barriers may limit, hinder and restrict professional outcomes which may erode and impact the self-esteem of women working in such spaces [24]. Such scholarship warrants the need to look into communal avenues of support seeking and resilience building such that women may seek refuge, guidance and find pathways to navigate professional crises and transitions [29]. In a parallel stream of scholarship, the use of social media as a potent vessel for support on a diverse spectrum of issues has been highlighted. Constructivist frameworks in this regard have surmised on how an emergent structure arises amidst the unregimented patterns of engagement, creating norms and values of these spaces and shaping the schema of participation and action adhered by such communities [23, 34]. Thus, the collective identity and structures of affinity such communities cultivate plays a pivotal role in the type of experiential sharing, learning and support mechanisms nurtured and offered [19]. Studies have shown how the collective identity created and sustained in these spaces, may drive the purpose they serve, for example, some may retain a more lightweight model of attachment serving primarily informational needs, while others may in addition create a community of support, empowerment and care, rooted in a more interpersonal model of trust and communal devotion [32]. Another burgeoning arc of investigation in this regard has pointed to how such channels can act as avenues of refuge by offering safe spaces to discuss and disclose a variety of stigmatized topics [20]. In this regard, the social capital curated and maintained in these spaces can be critical for the nature of narrative agency, advocacy and visions of empowerment nurtured [31]. Thus, online communities themselves can help to nurture and embolden acts of subversion that may help to galvanize a collective purpose aimed at tackling and dismantling structures of systemic and symbolic oppression [35]. Hence, the goal of my proposed work, is to connect these different avenues of scholarship and explore the way in which online communities can help to provide occasions for resilience building, empowerment and support for women as they navigate their careers, especially in the technical domain.

1.2 Research framing

Marrying the separate yet interconnected strands of literature, the goals of my dissertation work is to conduct two layers of investigation. The first will involve a mixed methods comparative analysis using content analysis, network analysis and complimentary linguistic analysis of three professional development subreddits (*r/cscareerquestions* (*r/csc*), *r/careerwomen* (*r/cw*), *r/girlsgonewired* (*r/ggw*)). This investigation is motivated by the fact that one of the key aims of my work is to investigate how online channels through group driven informal practices, can provide support, care and collective empowerment. Further, these Reddit forums have been selected as prior scholarship has shown the effectiveness of Reddit in acting as safe spaces for nurturing community driven disclosure patterns and as a space for extended discussion [1]. While the above exploration, can provide us an understanding of *what* type of conversations manifest, thereby giving a proxy of the collective structure the community establishes, there is also an equal

need to explicate *how* and *why* these community processes arise. To do so, through qualitative explorations, particularly through speculative design, the goal will be to uncover the way in which community members capture the values, norms and goals of these spaces and etch them into the way in which they participate and engage [34]. Thus, the central research questions that shall drive my explorations include:

- *As indicated by the content themes, psychosocial cues and emergent social structure, what kind of online community practice emerges in the forums of interest and how do the three forums differ in the nature of habitus they engender and the type of support and resilience strategies they nurture?*
- *What factors may shape the way in which women participate and seek support in such forums?*
- *In what ways might the future design and community maintenance practices of such spaces be envisioned to boost the collective efficacy of such channels?*

1.3 Methodological approach

Data from the three subreddits of interest will be collected using a mixture of existing databases on Reddit data (PSAW) and python API wrappers (PRAW). To conduct this investigation, a combined methodological panel of content analysis, network analysis and linguistic analysis will be employed. The motivation for using these different analysis frameworks stems from the fact that given the unstructured patterns of contribution in online domains, triangulating insights from multiple analytic tools can offer a deeper and richer insight into the practices and normative conventions of these virtual spaces [17]. Since textual conversations form the heart of the discourse patterns in these online channels, content analysis helps to gain insights into the type of conversational themes that drive the community dynamics of these spaces [28]. To conduct the content analysis, a thematic coding approaching, rooted in grounded theory will be followed to finalize the content categories present in the data [18]. In order to understand the latent psychosocial attributes that drive the conversational dynamics of these spaces, LIWC (Linguistic Inquiry and Word Count tool), one of the most widely used text analysis tools will be used [6]. While the content analysis and the added dimension of linguistic analysis can provide insights into the disposition of community members which manifests as their conversational traces, social network analysis (SNA) can provide an understanding of the type of positions or the way in which dyadic interactions carve out and are carved through the practices of these communities. In this case, the social network structure, represents each individual actor in the community as an independent node and the edges shall represent the interactivity or the commenting characteristics of these actors. Finally, to conduct the design studies, a specific kind of design fiction approach called futuristic autobiographies (FABs) shall be employed to elicit the values, norms and goals of these online spaces as captured through the eyes of community members, placing them as the central characters of the narrative scenarios presented [8]. This type of combined analysis allows us to survey, compare and ideate about the existing and future expansion of the social landscape as it exists in the virtual domain through the dimensions of community-building, resilience, and social support [4].

1.4 Preliminary Results

So far, I have completed a preliminary investigation of these three spaces with about 500 total conversational units from each of the communities. As a result of the content analysis, a set of six themes have been found. These themes cover a diverse spectrum of topics. While some are oriented towards providing more instrumental guidance regarding ways to develop knowledge and expertise, others focus more on providing support and care to those who face multiple forms of socio-cultural struggles and constraints. Such a cross-section of themes is further complemented by preliminary insights from the SNA analysis. The investigation of the social network constructed using the sample of conversations extracted from each of the forums, highlights a different forum of communal structure and orientation of these communities. While the general forum on technical careers appears to have many fragmented communities, the women centric forums have larger connected components indicated a stronger community focus in these spaces. These inferences are also supported by other network metrics which highlight that the strongest communal affinity manifests in the women-centric technical forum, followed by the career forum focused on women and finally the generic technical forum ranks the lowest in this regard. From these insights, we glean insights into the way in which the gendered habitus, organizational habitus and the emergent habitus of online community practice intersect, creating a unique perspective, goals and values for each such space. Specifically, with regard to the women-centric forums, similar to perspectives drawn from offline contexts [25], it is evident that these online enclaves help bolster collective resistance stances and discussions as a form of everyday resilience. The anecdotal sharing based on personal experiences depicts a strong sense of fellow feeling, expressions of peer parity and overall commitment to the community [29].

Yet, the habitus in each of the forums depicts a unique form of collective purpose and mission. The generic technical forum represents a form of virtual enclave that acts as an informational hub, a weak tie model of attachment, whereas the other two forums demonstrate a strong sense of community, collective action and shared understanding of trauma and struggles faced by women in navigating their career ventures [12]. Between the two women centric forums, an interesting perspective emerges. The linguistic markers signify a greater moral orientation and collective curiosity in that of r/girlsgonewired (the technically oriented women centric forum). This is further backed by complementary evidences from content and network analysis, that show that r/girlsgonewired encourages provocative discussions centered on representation, recognition and the need for enhancing inclusivity in technical domains. This perspective adds a dimension of relational orientation among the members of r/girlsgonewired that nurtures the development of communal solidarity centered on care, moral and ideological deliberation and interpersonal trust, modeling an emergent ethic of care in such channels [14]. In summary, the generic women centric forum (r/careerwomen) represents an alliance of professional women to network, inspire, unite and provides a ground for community building and support. While a similar spirit exists in the women centric forum with a technical focus (r/girlsgonewired), it essentially exhibits an additional dimension of opposing the habitus of exclusion and cultural oppression through collective community

endeavors and narrative crusades giving such online collectives a unique purpose, vision and voice centered on empowerment and resistance.

1.5 Expected contributions

Theoretically, this study contributes by conceptualizing a theoretical amalgam with the concept of the gender habitus serving as the primary lens, combined with frameworks on resilience building to navigate every day crises for historically underrepresented communities. On a practical and empirical level of contribution, these themes indicate ways in which such communities offer extended mentoring and can provide augmented support [10]. Such resources can also provide extended peer support in times of crisis which can disrupt standardized work and learning practices [2]. These investigations can also contribute and add to the growing body of literature in the context of career research in HCI [33], which is another stream of scholarship which needs more rigorous exploration. My dissertation work can add to this theme by providing ways to conceptualize systems that can act as informal pathways for support, collaboration and guidance beyond traditional institutionalized resources.

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